BRT Safety Filtering for LLM-Generated Orbital Maneuver Plans

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***Abstract*—Large language models (LLMs) are increasingly being explored for autonomous mission planning, offering the ability to translate natural language objectives into actionable maneuver sequences. However, LLMs have no internal model of orbital mechanics or physical safety constraints, and can produce plans that violate keep-out zones or lead to conjunction with a target object. This paper proposes a safety filtering pipeline in which a precomputed Backward Reachable Tube (BRT), derived via Hamilton-Jacobi (HJ) reachability analysis over Clohessy-Wiltshire relative motion dynamics, intercepts unsafe LLM-generated maneuver commands and replaces them with minimum-norm safe alternatives. We describe the problem formulation, proposed approach, and evaluation plan, with full implementation and results to follow.**

***Index Terms*—Hamilton-Jacobi reachability, backward reachable tube, safety filtering, large language models, orbital proximity operations, Clohessy-Wiltshire equations.**

I. Introduction and Motivation

AUTONOMOUS on-board mission planning is an increasingly important capability for spacecraft operating in environments where ground-in-the-loop latency is prohibitive. In deep-space missions, round-trip communication delays of several minutes to hours make real-time ground intervention during time-critical proximity operations infeasible. In congested low Earth orbit (LEO), intermittent contact windows mean that a spacecraft may need to react to a conjunction event without ground oversight. In both cases, autonomous planners that can generate and execute maneuver sequences without human approval are desirable.

Large language models (LLMs) have recently attracted interest as a component of autonomous planning systems. Their ability to interpret natural language mission objectives and produce structured command sequences offers a flexible interface between operator intent and spacecraft actuation [6]. An operator could, in principle, specify a proximity operations objective in plain English — "approach the target from the anti-velocity direction, maintaining at least 50 meters separation" — and an LLM would produce a corresponding sequence of ΔV maneuver commands.

The critical problem is that LLMs have no internal representation of orbital mechanics, physical safety constraints, or the keep-out zones (KOZs) that define safe proximity operations. A plan that is linguistically coherent may drive the chaser spacecraft directly into the exclusion region surrounding the target. Without a formal safety layer, the gap between *semantic planning* (what the mission intends) and *formal safety* (what physics allows) remains unaddressed.

Existing work on LLM safety for robotic systems has proposed reachability-based runtime monitors for general robot navigation [5]. Formal methods have also been applied to spacecraft mission planning via Markov decision process (MDP) frameworks [10]. However, to the best of our knowledge, no existing work applies Hamilton-Jacobi reachability as a runtime safety filter specifically for LLM-generated orbital maneuver plans, where the BRT simultaneously identifies unsafe states and provides a principled minimum-cost correction.

This paper addresses that gap. We propose a pipeline in which an LLM generates proximity maneuver plans from natural language mission specifications, and a precomputed BRT serves as a hard safety filter. At each timestep, the filter checks whether the planned state lies within the BRT; if so, the unsafe command is replaced with the minimum-norm projection onto the BRT boundary. The outcome is a formal safety guarantee over the closed-loop LLM-plus-filter system, independent of the LLM's internal reasoning.

II. Related Work

A. Hamilton-Jacobi Reachability for Spacecraft Safety

Hamilton-Jacobi reachability analysis provides a principled framework for computing the Backward Reachable Tube (BRT) — the set of all states from which a system will inevitably enter a failure set, regardless of control input [1]. For proximity operations, this framework has been applied to compute safe regions around target spacecraft under Clohessy-Wiltshire dynamics [4]. The value function $V(x, t)$ satisfies a Hamilton-Jacobi partial differential equation (PDE) solved backward in time, and the BRT boundary $\partial\mathcal{B}$ represents the "last safe moment" at which corrective action can prevent a violation. Grid-based solvers such as the `hj_reachability` toolbox [2] have made these computations tractable for systems up to approximately 6 dimensions without GPU requirements.

B. LLMs for Autonomous Mission Planning

Recent work has demonstrated the potential of LLMs for task planning in robotics and aerospace contexts [6]. LLMs can decompose high-level natural language goals into structured action sequences and have been applied to satellite scheduling, task prioritization, and trajectory description. However, their probabilistic, text-based reasoning provides no guarantee of physical feasibility or constraint satisfaction. Work on *constrained* LLM planning has explored post-hoc verification via temporal logic [7] and SMT-based barrier certificate synthesis [8], but these approaches do not provide runtime-safe control inputs for dynamical systems.

C. Safety Filtering and CBFs for Spacecraft

Control Barrier Functions (CBFs) have been applied to spacecraft proximity operations to enforce keep-out zones via quadratic programming at each control step [9]. While CBF-based filters can provide safety guarantees for continuous-control systems, they require a known, differentiable barrier function and may be infeasible when the control set is finite (e.g., on/off thrusters). HJ reachability offers a complementary approach that handles finite control sets naturally and is robust to worst-case disturbances, at the cost of offline precomputation. The proposed pipeline adopts reachability-based filtering rather than CBF-based filtering for these reasons.

D. Safety Under Localization Uncertainty

A related line of work addresses safety when state measurements are uncertain. Robust CBFs [11] and stochastic reachability [12] have been proposed to inflate safety margins as a function of state estimation error. While the present work assumes perfect state knowledge, these formulations suggest natural extensions to scenarios where the chaser state is only partially observed.

III. Problem Formulation

A. Relative Motion Dynamics

We model the motion of a chaser spacecraft relative to a target in a near-circular reference orbit using the Clohessy-Wiltshire (CW) equations [3]. The state vector is

$$x = [r_x,\ r_y,\ r_z,\ \dot{r}_x,\ \dot{r}_y,\ \dot{r}_z]^\top \in \mathbb{R}^6, \tag{1}$$

where (r_x, r_y, r_z) denotes the relative position in the radial, in-track, and cross-track directions, and $(\dot{r}_x, \dot{r}_y, \dot{r}_z)$ the corresponding relative velocities. The linearized dynamics are

$$\dot{x} = Ax + Bu + w, \tag{2}$$

where n is the mean motion of the reference orbit,

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \otimes I_3, \tag{3}$$

$u \in \mathcal{U} \subset \mathbb{R}^3$ is the applied ΔV control input, and $w \sim \mathcal{N}(0, Q)$ is process noise representing unmodeled dynamics.

B. Keep-Out Zone

The target spacecraft is surrounded by an ellipsoidal keep-out zone

$$\mathcal{F} = \left\{ x \in \mathbb{R}^6 : \frac{r_x^2}{a^2} + \frac{r_y^2}{b^2} + \frac{r_z^2}{c^2} \le 1 \right\}, \tag{4}$$

with semi-axes $a, b, c > 0$ parameterizing the exclusion region. This constitutes the *failure set*: any state in $\mathcal{F}$ corresponds to a conjunction (collision or proximity violation).

C. Backward Reachable Tube

The Backward Reachable Tube $\mathcal{B}(t)$ is defined as the set of all states from which the system will enter $\mathcal{F}$ within time horizon $[t, T]$, regardless of the control applied:

$$\mathcal{B}(t) = \{x_0 : \exists\, \tau \in [t, T],\ x(\tau) \in \mathcal{F} \text{ for all } u(\cdot) \in \mathcal{U}\}\,. \tag{5}$$

The BRT is characterized by the viscosity solution $V : \mathbb{R}^6 \times [0, T] \to \mathbb{R}$ of the Hamilton-Jacobi-Isaacs PDE

$$\frac{\partial V}{\partial t} + \min\left(0,\ H(x, \nabla_x V)\right) = 0, \quad V(x, T) = l(x), \tag{6}$$

where $l(x)$ is a signed distance function to $\mathcal{F}$ and the Hamiltonian is

$$H(x, p) = \min_{u \in \mathcal{U}}\ p^\top f(x, u). \tag{7}$$

The BRT at time t is then $\mathcal{B}(t) = \{x : V(x, t) \le 0\}$.

D. LLM Maneuver Plan

Let $\pi_{\text{LLM}} = \{(k, \Delta V_k)\}_{k=0}^{N}$ denote a maneuver plan produced by an LLM, where k indexes discrete timesteps and $\Delta V_k \in \mathbb{R}^3$ is the commanded velocity increment at step k. The plan is generated by prompting the LLM with a natural language mission specification $\mathcal{M}$ (e.g., approach direction, target range, urgency) and requesting a JSON-formatted output.

E. Safety Filtering Problem

Given a state x_k at timestep k and an LLM-commanded input ΔV_k, the safety filter must produce a safe input ΔV_k^* such that the resulting state x_{k+1} lies outside $\mathcal{B}(t_{k+1})$, while minimizing the correction cost $\|\Delta V_k^* - \Delta V_k\|$. Formally,

$$\Delta V_k^* = \arg\min_{\Delta V} \|\Delta V - \Delta V_k\| \quad \text{s.t.} \quad V(f(x_k, \Delta V),\, t_{k+1}) > 0. \tag{8}$$

If $V(f(x_k, \Delta V_k), t_{k+1}) > 0$, the LLM command is already safe and $\Delta V_k^* = \Delta V_k$. Otherwise, ΔV_k^* is the minimum-norm projection of ΔV_k onto the safe set boundary.

IV. Proposed Approach

A. BRT Precomputation

The BRT is computed offline using the `hj_reachability` grid-based solver [2] over the 6-dimensional CW state space. The failure set $\mathcal{F}$ is defined as the ellipsoidal KOZ described above. The solver propagates the HJ PDE backward over a time horizon T sufficient to cover the proximity approach scenario (typically 1–2 orbital periods). The resulting value function $V(x, t)$ is stored as a grid and queried at runtime via trilinear interpolation.

B. LLM Prompting Pipeline

We use a frontier LLM accessed via the Anthropic API. Each prompt provides: (1) a description of the proximity operations scenario including the target location and KOZ parameters; (2) the current chaser state; and (3) a natural language mission specification $\mathcal{M}$. The LLM is instructed to return a maneuver plan as a JSON object of the form:

```
{
  "plan": [
    {"timestep": 0, "dv": [ax, ay, az]},
    {"timestep": 1, "dv": [bx, by, bz]},
    ...
  ]
}
```

Mission specifications are varied across approach angle (radial, in-track, anti-velocity), urgency framing ("immediately," "when safe"), and initial range to the KOZ boundary. A corpus of 50–100 plans is generated.

C. Runtime Safety Filter

At each timestep k, the filter evaluates $V(x_{k+1}^{\mathrm{LLM}}, t_{k+1})$ where $x_{k+1}^{\mathrm{LLM}} = f(x_k, \Delta V_k)$ is the state predicted by applying the LLM command. If $V \leq 0$, the state is unsafe and the filter solves the minimum-norm correction problem. The correction is implemented as a line search from ΔV_k toward the nearest safe command, evaluated against the BRT value function.

D. Evaluation and Baselines

Three conditions are evaluated over the full plan corpus:

1) **No filter**: LLM plans executed without intervention.
2) **BRT filter**: LLM plans with the proposed safety filter applied.
3) **Rule-based planner**: A fixed radial burn sequence with no LLM component, serving as a non-LLM safety anchor.

Primary metrics are: (a) *interception rate* — fraction of plans requiring at least one filter intervention; (b) ΔV *overhead* — total correction cost averaged over the plan corpus; and (c) *mission success rate* — fraction of plans in which the chaser reaches within threshold distance of the target without entering $\mathcal{F}$, before and after filtering.

V. Preliminary Results

No implementation results are available at this stage. The CW dynamics model and KOZ geometry have been specified. BRT computation, LLM pipeline, and filter implementation are in progress per the milestones below.

VI. Milestones and Work Plan

- **Week 6** (complete): Project scoping, literature review, dynamics model specified, KOZ geometry defined, sanity check against known conjunction geometry.
- **Week 7**: Implement BRT computation using `hj_reachability`. Verify BRT boundary geometry. Implement safety filter (BRT query + minimum-norm correction).
- **Week 8**: Build LLM prompting pipeline. Generate full plan corpus (50–100 plans) across varied mission specs. Run filter over all plans.
- **Week 9**: Evaluate all metrics. Run baseline comparisons. Stress test edge cases (short-horizon plans, ambiguous prompts, near-boundary initial conditions).
- **Week 10**: End-to-end debugging and final testing. Write final report and prepare presentation.

VII. Conclusion

We have described a pipeline for enforcing formal safety guarantees on LLM-generated orbital maneuver plans using Hamilton-Jacobi reachability. The mathematical formulation, approach, and evaluation plan have been established. Implementation is underway and full results will be reported in the final paper.

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